NATIONAL INSTITUTE OF TECHNOLOGY: TIRUCHIRAPPALLI: 620 015

MECHANICAL ENGINEERING DEPARTMENT.

Cycle Test - I

Branch: Instrumentation and Control Engineering **Semester: I**

MEIR 11 – Basics of Mechanical Engineering.

Time: 60 min

Date: 13.10.2026

1. How to increase the volumetric efficiency of IC engine? (1)
2. What happens after the compression stroke in four stroke diesel engine? How is it different from petrol engine? (3)
3. What is meant by scavenging in two stroke engine. (1)
4. What are the four main components of Vapour compression refrigeration system and explain their functions. Draw a schematic for this system. (4)
5. A household refrigerator that has a power input of 450 W and COP of 3 is to cool six watermelons 10 kg each to 8°C. If the fruits are initially at 20°C, determine how long it will take for the refrigerator to cool them. Assume the specific heat of watermelon as 4.2 kJ/kg.K. (2)
6. What is meant by superheated steam and wet steam? (1)
7. Calculate piston speed in m/s for IC engine running at 2500 rpm. The stroke to bore ratio is 1.5 and the bore is 50 mm. (1)
8. Explain Rankine cycle with the help of schematic diagram and T-s plot. (4)
9. What is the purpose of Economiser in thermal power plant? (1)
10. What is meant by Fly-Ash? (1)
11. How the cooling water used for condenser is cooled in cooling tower? (1)